Constrained Generative models for Shape Deformation and Reduced Order Modelling

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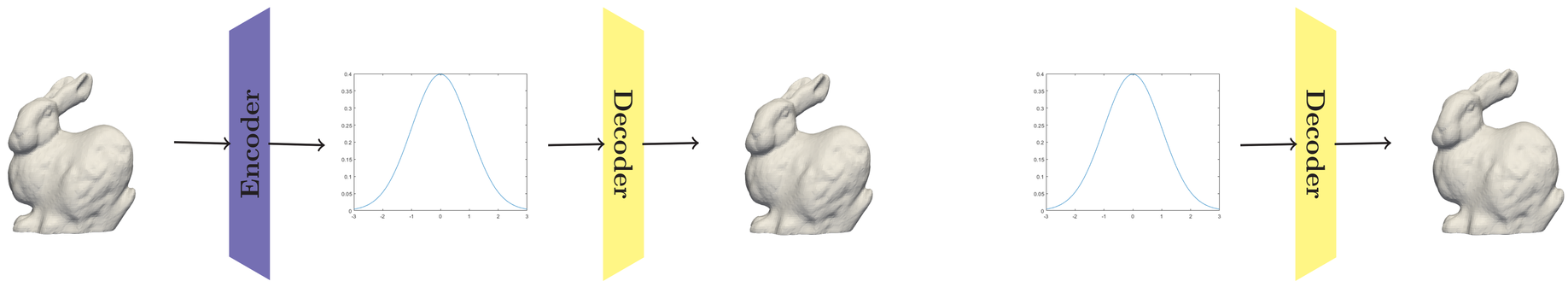
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Introduction

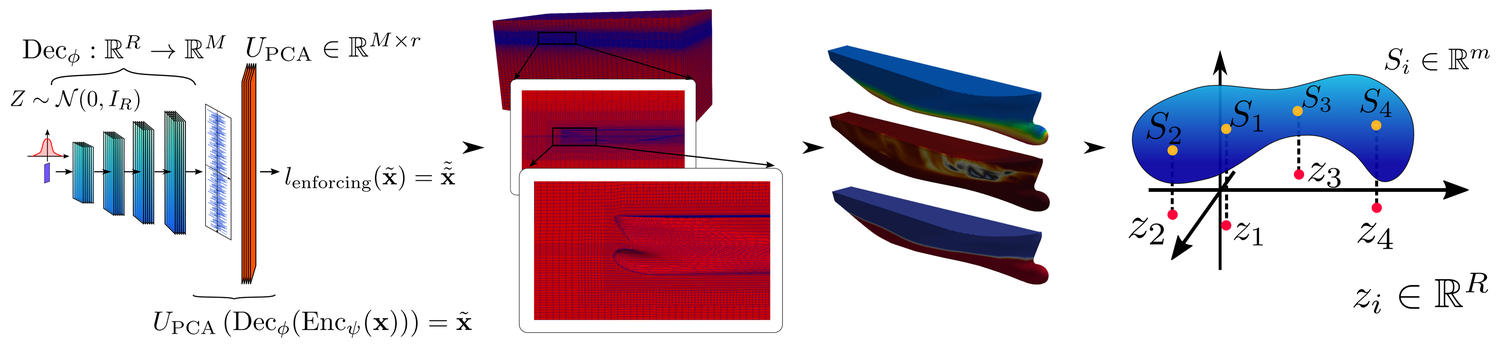
We study **generative models** for shape optimization of complex geometries with a large number of parameters; the objective is also to reduce the number of relevant geometrical parameters, for example for modeling naval hulls, and creating new artificial geometries similar to real data, as there are non-generative techniques for creating new real geometries which respect some constraints, for having fixed volume, (see the package PyGem [1] developed at SISSA) but using them can be costly. The real geometries are parametrized by a lesser number of parameters, which in turn increases the performance of reduced order models.

Generative models for reduction in parameter space

This study employs generative models to learn the distribution of complex 3D shapes while simultaneously achieving parameter reduction. Specifically, the reduced parameter space corresponds to the latent space of the generative model. Two distinct approaches were adopted in our works. In the first approach, meshes of fixed topology that respect a set of invariants are used for training generative models, which are augmented with Principal Component Analysis (PCA) and with a final projection step restoring exact constraint satisfaction, and have a low-dimensional latent space. The second approach forgoes PCA by inputting both the latent code and point coordinates into the generative network decoder to handle varying mesh topologies.

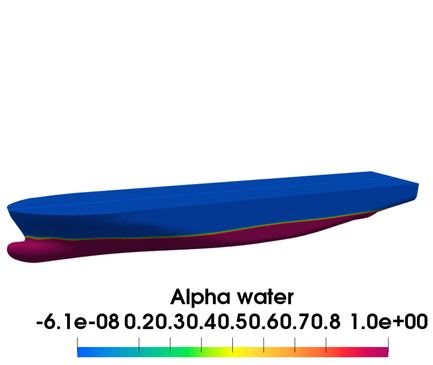


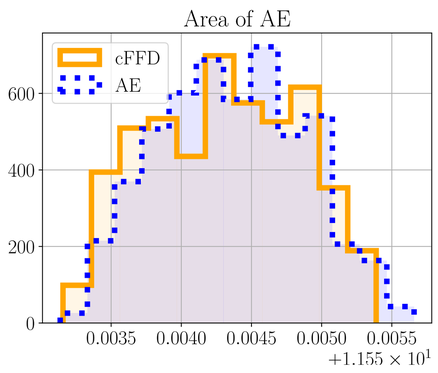
Constrained Generative Models for Shape Deformation on Fixed Topology Meshes

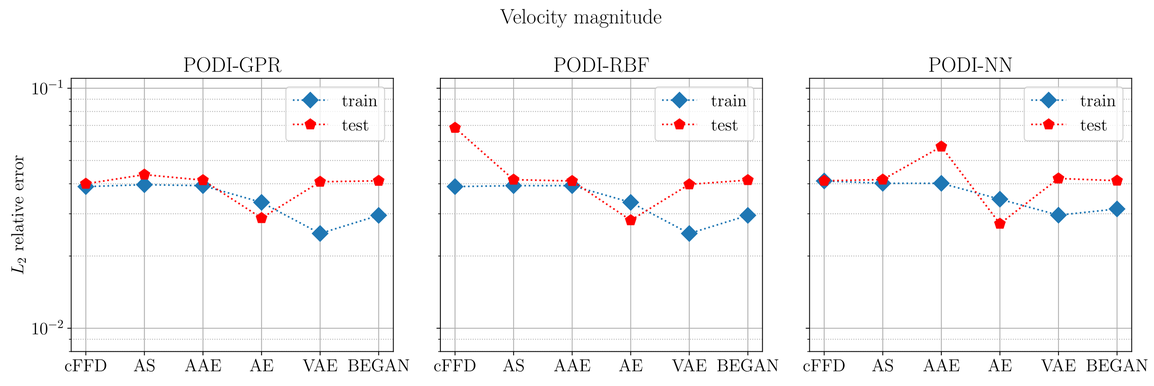


sampling from cGMs → *mesh generation: RBF interpolation* → *MOR: snapshots collection* → *PODI: design of response surfaces*

We study two test case: one of them is a multifluid Interfoam simulation on the DTCHull, with the bulb deformed using Constrained Free Form Deformation with fixed volume (figure on the left). As generative models we adopt Variational Autoencoders, Adversarial Autoencoders, Classic Autoencoders and Boundary Equilibrium Generative Adversarial Networks. We validate our model by checking the distribution (on the center) and the performance of reduced order models on quantity of interest (on the right).



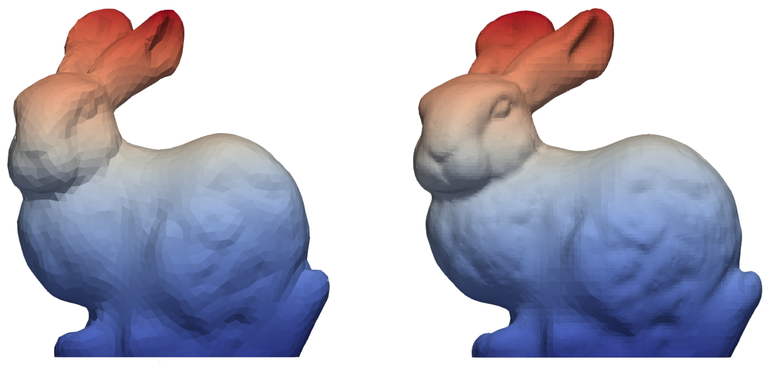


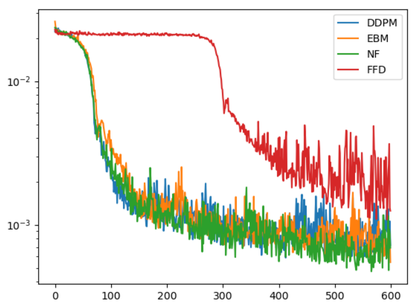


The other test case is the deformation of a Stanford Bunny with fixed barycenter. We refer to [2] for the details.
As future work, we could try to improve our generative models with the adoption of Graph Neural Networks.

Constrained Generative Models for Shape Deformation on Arbitrary Topology Meshes

We modify the deep generative models to have as inputs a latent code and a point which belongs to a reference mesh, and which outputs a point on a deformed mesh [3]. By learning a PPINN on the reduced latent space, the loss convergence is faster, and the curse of dimensionality is substantially alleviated.
As generative models we adopt Normalizing Flows, Energy Based Models, Denoising Diffusion Probabilistic Models, coupled with a deterministic Autoencoder. On the left, a coarse mesh used for training, on the center dense mesh user for testing, on the right the testing error as the number of epochs increases.





Bibliography and Software References

[1] Pygem, `https://github.com/mathLab/PyGeM`

[2] Padula, G., Romor, F., Stabile, G. & Rozza, G. Generative models for the deformation of industrial shapes with linear geometric constraints: Model order and parameter space reductions. *Computer Methods In Applied Mechanics And Engineering*, 423, pp. 116823 (2024).

[3] Padula, G. & Rozza, G. Generative Models for Parameter Space Reduction applied to Reduced Order Modelling *arXiv*, 2506.09721 (2025).